

X3 ULTRA Battery Selection



X3 Ultra Inverter Information

The X3 Ultra has two separate battery ports for two strings of batteries, here's some data on the inverter's battery port.

Table 1 Battery Port Data

Inverter Type	X3-ULT-15K/19.9K/20K/25K/30K
Battery voltage range [V]	180*-800
Max. charge / Discharge current [A]	30 × 2
*The voltage range of the battery port will be reduced shortly.	

Compatibility

The X3 Ultra is available with the following 5 types of batteries. More types of batteries are being adapted, please stay tuned.



Figure 1 Batteries for X3 Ultra

Battery Type	Each Port	Battery Heating	Battery Extend	Advantages	Notes
HS25/36	4~13	Yes	TCBOX-70 (Supports 3 strings)	1. Attractive Design 2. Easy installation, plug and play 3. Large capacity for system expansion. 4. Large charge and discharge current, which can accelerate the charging speed.	The maximum quantity of one tower is 8pcs. If the number of HS batteries on one terminal ≥ 9pcs, a Series Box is required. The battery heating function is under development.

HS51	3~13	Yes	TCBOX-70 (Supports 3 strings)	1. Attractive design. 2. Easy installation, plug - and - play. 3. Large capacity for system expansion. 4. Large charge and discharge current, which can accelerate the charging speed.	The maximum quantity of one tower is 8pcs. If the number of HS batteries on one terminal ≥ 8 pcs, a Series Box is required.
HR25/36	4~13	No	TCBOX-70 (Supports 3 strings)	1. Industrial Style Design 2. Easy installation 3. Large capacity for system expansion.	
T58	2~4	No	BMS Parallel Box G2 (Supports 2 strings)	1. Large single battery capacity	
T30	2~4	Yes	BMS Parallel Box G2 (Supports 2 strings)	1. Modular stacking design, easy installation	The battery heating function for BMS Parallel Box II G2 with T30 is under development.

Table 2 Battery Comparison

Single and Dual Battery Port Solution

Battery Solution

1. HS(HR)25/36 and HS51 1 in 1 Solution

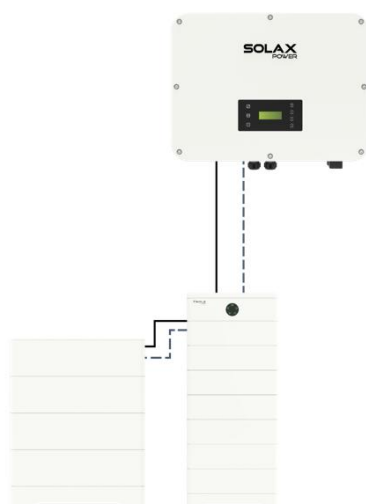


Figure 2 X3 Ultra+One BAT port

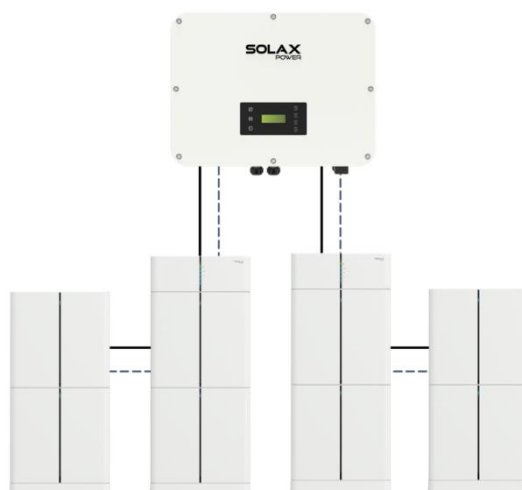


Figure 3 X3 Ultra+two BAT port

	T-BAT-SYS-HV-S2.5	T-BAT-SYS-HV-S3.6	TSYS-HS51
Adaptable Quantity	$(4-13) \times 2$ pcs	$(4-13) \times 2$ pcs	$(3-13) \times 2$ pcs
Total Capacity[kWh]	$(10.2-33.2) \times 2$	$(14.7-47.9) \times 2$	$(15.3-66.5) \times 2$
Max. charge / discharge current [A]	30×2	30×2	30×2
Nominal voltage [V]	204.8-665.6	204.8-665.6	153.6-665.6
Power [kW]	$(6.1-19.9) \times 2$	$(6.1-19.9) \times 2$	$(6.1-19.9) \times 2$

Table 3 Basic Data for 1 in 1 Solution

2. HS2.5/3.6 and HS51 2 in 1 Solution

The X3 ultra supports paralleling two battery ports together via a 2-in-1 paralleling cable, which allows the battery to pass a higher current, allowing the battery to be charged and discharged to its full matching power.

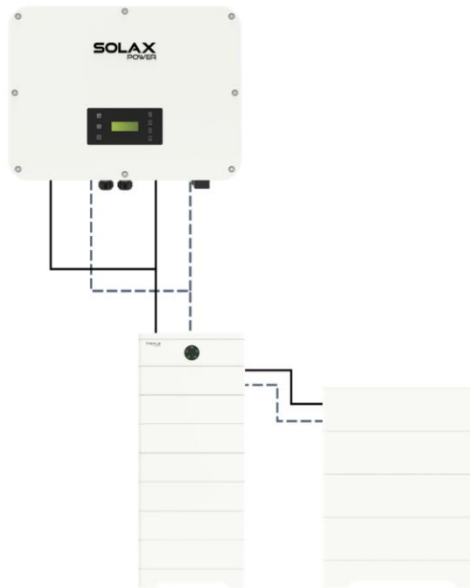


Figure 4 X3 Ultra + 2-in-1 cable+HS 2.5/3.6

	T-BAT-SYS-HV-S2.5	T-BAT-SYS-HV-S3.6	TSYS-HS51
Adaptable Quantity	4-13 pcs	4-13 pcs	3-13 pcs
Total Capacity[kWh]	10-33.2	14.7-47.9	15.3-66.5
Max. charge / discharge current [A]	45A	50A	60A
Nominal voltage [V]	204.8-665.6	204.8-665.6	153.6-665.6
Power [kW]	9.2-29.9	10.2-33.2	10.8-44.6

Table 4 Basic Data for HS25/36 2 in 1 Solution

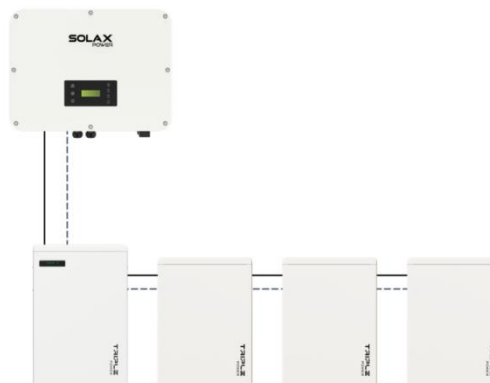


Figure 5 X3 Ultra + One BAT port

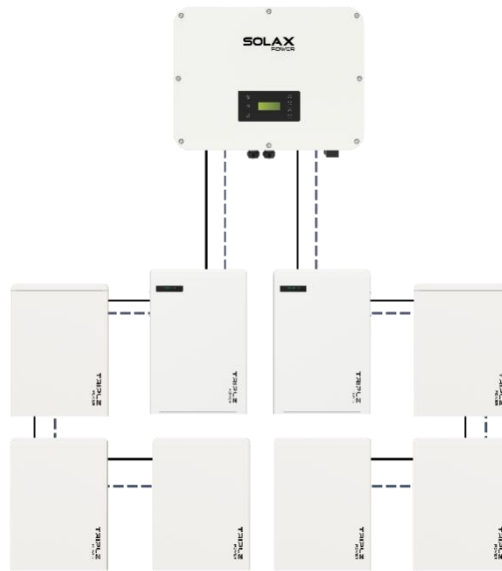


Figure 6 X3 Ultra + Two BAT port

	T-BAT H 5.8
Adaptable Quantity	$(2-4) \times 2$ pcs
Total Capacity[kWh]	$(11.5-23) \times 2$
Max. charge / discharge current [A]	30×2
Nominal voltage [V]	230.4-460.8
Power [kW]	$(6.9-13.8) \times 2$

Table 5 Basic Data for T58 Separate BAT Solution

T30 1 in 1 Solution

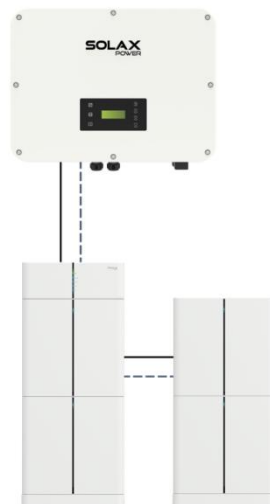


Figure 7 X3 Ultra + One BAT port

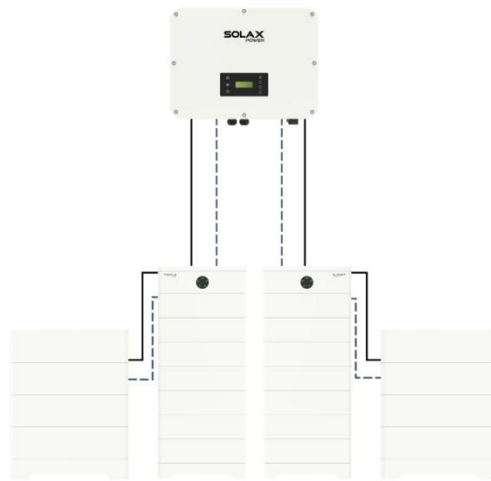


Figure 8 X3 Ultra + Two BAT port

	T-BAT H 3.0
Adaptable Quantity	$(2-4) \times 2$ pcs
Total Capacity[kWh]	$(6.1-12.3\text{kWh}) \times 2$
Max. charge / discharge current [A]	30×2
Nominal voltage [V]	204.8 -409.6
Power [kW]	$(6.1-12.3) \times 2$

Table 6 Basic Data for T30 Normal Solution

Battery Extend

X3 Ultra can realize battery extend by various accessories adaptable with different batteries, such as TCBOX-70 and BMS Parallel Box G2. And please note that the number of batteries in both strings needs to be the same.

1. HS(HR)2.5/3.6 and HS51 TC-Box-70 Solution

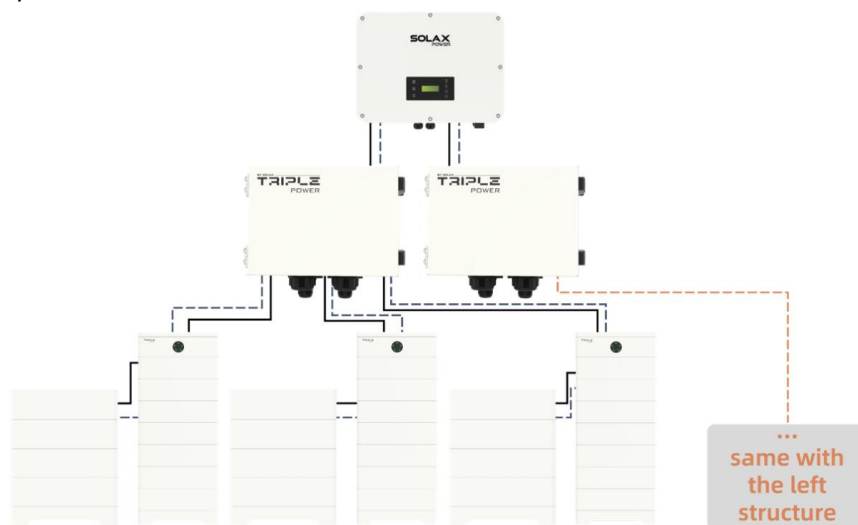


Figure 9 X3 Ultra + TC-Box-70+HS 2.5/3.6

	T-BAT-SYS-HV-S2.5	T-BAT-SYS-HV-S3.6	TSYS-HS51
Adaptable Quantity	$(4-13) \times 6$ pcs	$(4-13) \times 6$ pcs	$(3-13) \times 6$ pcs
Total Capacity[kWh]	$(10-32.5) \times 6$	$(14.7-47.9) \times 6$	$(15.3-66.5) \times 6$
Max. charge / discharge current [A]	30×2	30×2	30×2
Nominal voltage [V]	204.8-665.6	204.8-665.6	204.8-665.6
Power [kW]	$(6.1-19.9) \times 2$	$(6.1-19.9) \times 2$	$(6.1-19.9) \times 2$

Table 7 Basic Data for HS2.5/3.6/51 TC-Box-70 Solution

2. T58 BMS Parallel Box II G2 Solution

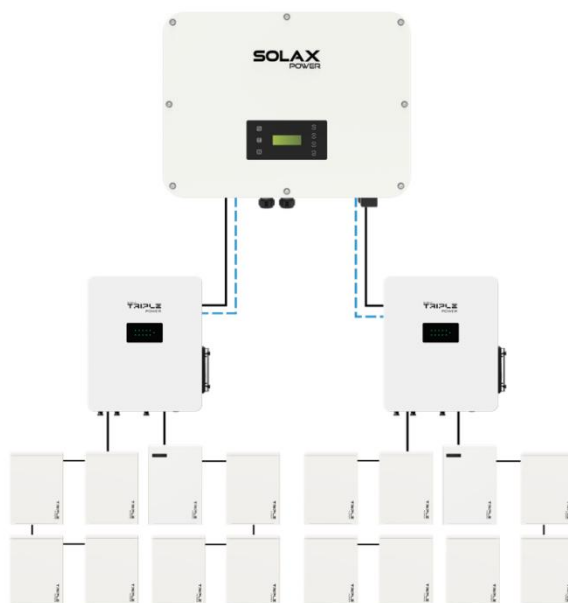


Figure 7 X3 Ultra + BMS Parallel Box II G2 +T58

	T-BAT H 5.8
Adaptable Quantity	$(4 \text{ or } 6 \text{ or } 8) \times 2$ pcs
Total Capacity[kWh]	$(23 \text{ or } 34.6 \text{ or } 46.1) \times 2$
Max. charge / discharge current [A]	30×2
Nominal voltage [V]	230.4 or 345.6 or 460.8
Power [kW]	$(6.9 \text{ or } 10.4 \text{ or } 13.8) \times 2$

Table 8 Basic Data for T58 BMS Parallel Box G2 Solution

3. T30 BMS Parallel Box II G2 Solution

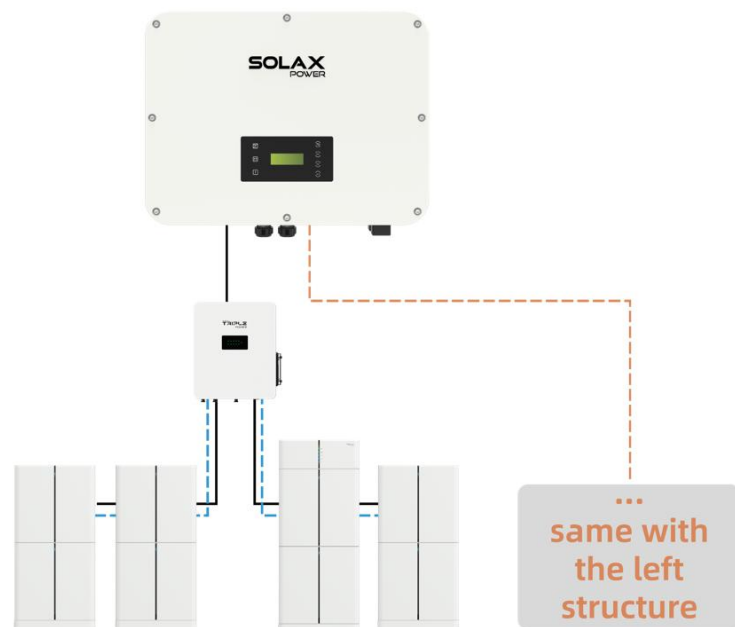


Figure 11 T30+BMS Parallel Box II G2

	T-BAT H 3.0
Adaptable Quantity	(4 or 6 or 8) × 2 pcs
Total Capacity[kWh]	(12.3-24.6) × 2[kWh]
Max. charge / discharge current [A]	30 × 2
Nominal voltage [V]	204.8 -409.6
Power [kW]	(6.1-12.2)× 2

Table 9 Basic Data for T30 BMS Parallel Box II G2 Solution

System and Battery extend

1. External Parallel(Up to 3pcs)

The X3-Ultra supports 3pcs internal switching parallel. Here is the data for each type of battery with their compatible parallel box.

Battery Type	HS(HR)25/36/51	T58	T30
Accessories	TCBOX-70	BMS Parallel Box G2	BMS Parallel Box G2
Quantity	6×13×3pcs	4×4×3pcs	4×4×3pcs
Capacity[kWh]	585/862.2/1197	276k	147.6

Table 10 Data for external parallel solution

2. External Parallel Solution(Up to 5pcs)

The X3-Ultra supports 5 pcs external switching parallel, and when extending the number of batteries in both strings needs to be the same.

① HS(HR)25/36/51

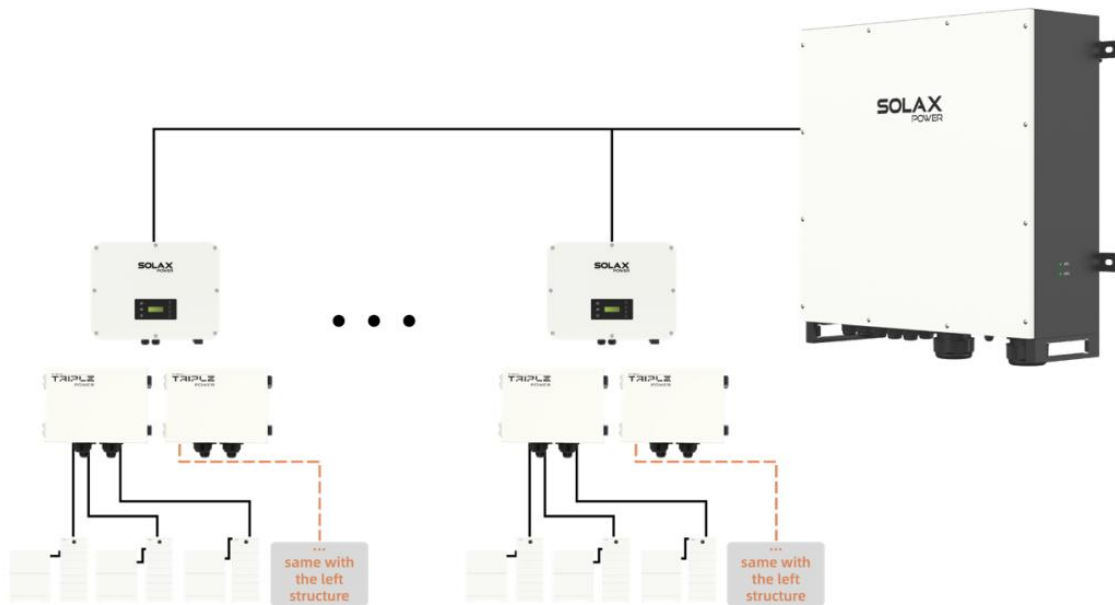


Figure 12 X3 Ultra + EPS Parallel Box G2.1-150K(adaptable with X3 Ultra)+ HS25/36

	T-BAT-SYS-HV-S2 .5	T-BAT-SYS-HV-S3.6	TSYS-HS51
Adaptable Quantity	$(4-13) \times 6 \times 5$ pcs	$(4-13) \times 6 \times 5$ pcs	$(3-13) \times 6 \times 5$ pcs
Total Capacity[kWh]	$(10-33.2) \times 6 \times 5$	$(14.7-47.9) \times 6 \times 5$	$(15.3-66.5.) \times 6 \times 5$
Max. charge / discharge current [A]	30×2	30×2	30×2
Nominal voltage [V]	204.8-665.6	204.8-665.6	204.8-665.6
Power [kW]	$(6.1-19.9) \times 6 \times 5$	$(6.1-19.9) \times 6 \times 5$	$(6.1-19.9) \times 6 \times 5$

Table 11 Basic Data for External Switching Parallel Solution

②T58

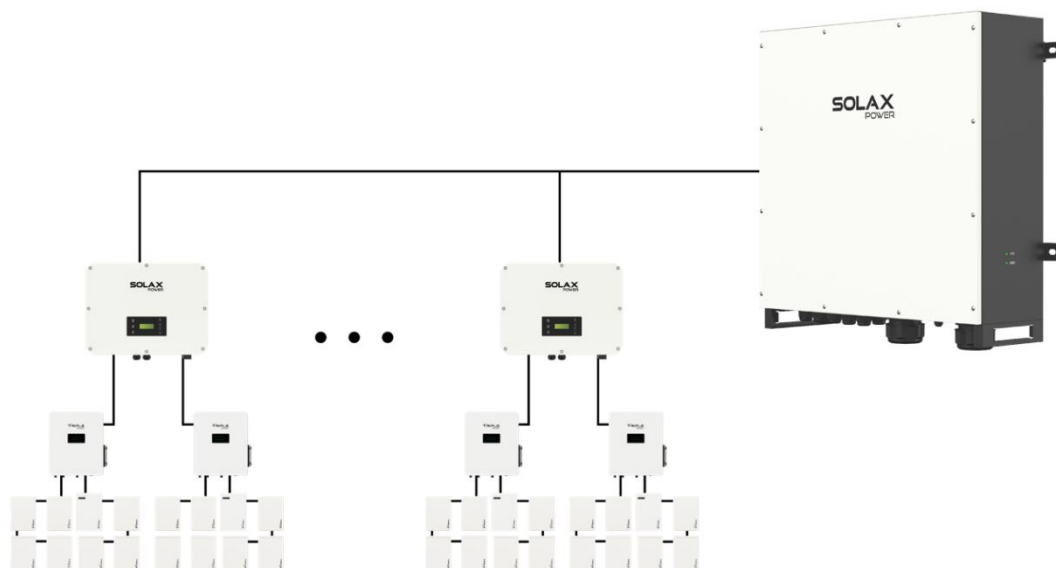


Figure 13

X3 Ultra + BMS Parallel Box G2 +
EPS Parallel Box G2.1-150K(adaptable with X3 Ultra)+T58

	T-BAT H 5.8
Adaptable Quantity	$(4 \text{ or } 6 \text{ or } 8) \times 2 \times 5 \text{ pcs}$
Total Capacity[kWh]	$(23 \text{ or } 34.6 \text{ or } 46.1) \times 2 \times 5$
Max. charge / discharge current [A]	30×2
Nominal voltage [V]	230.4 or 345.6 or 460.8
Power [kW]	$(6.9 \text{ or } 10.4 \text{ or } 13.8) \times 2 \times 5$

Table 12 Basic Data for External switching parallel Solution

③T30

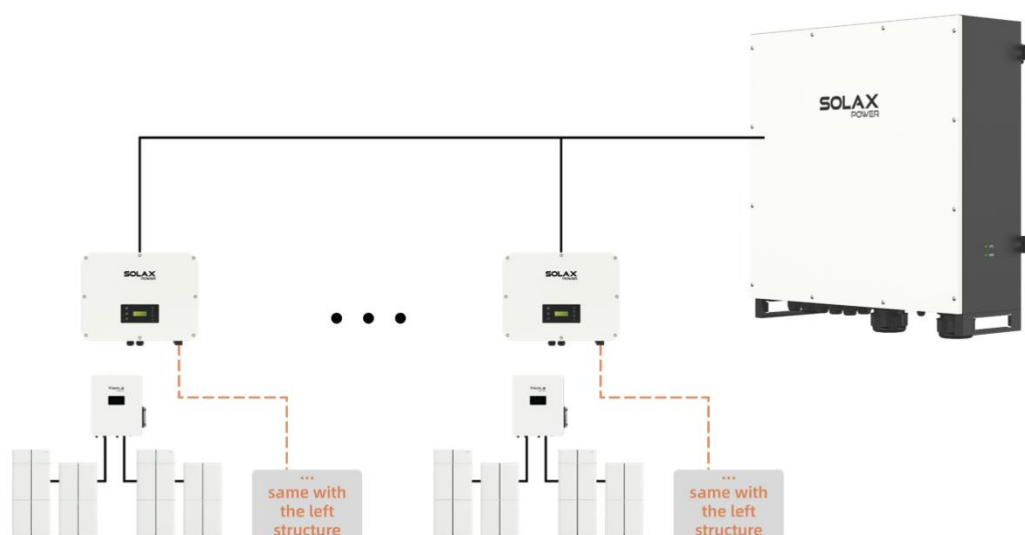


Figure 14 X3 Ultra + BMS Parallel Box G2 +EPS Parallel Box G2.1-150K+T30

	T-BAT H 3.0
Adaptable Quantity	$(4 \text{ or } 6 \text{ or } 8) \times 2 \times 5 \text{ pcs}$
Total Capacity[kWh]	$(12.3-24.6) \times 2 \times 5$
Max. charge / discharge current [A]	30×2
Nominal voltage [V]	204.8 - 409.6
Power [kW]	$(6.1-12.2) \times 2 \times 5$

Table 13 Basic Data for External switching parallel Solution

3. X3-Ultra external-switching 10 pcs parallel programme is being planned, please stay tuned!